

My Project

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1 File Index	1
1.1 File List	1
2 File Documentation	3
2.1 NaiveBayesClassifier.py File Reference	3
2.1.1 Detailed Description	4
2.1.2 Function Documentation	4
2.1.2.1 calculate_dice_coefficient()	4
2.1.2.2 calculate_iou()	4
2.1.2.3 evaluate_segmentation()	5
2.1.2.4 extract_features_with_coords()	5
2.1.2.5 main()	6
2.1.2.6 predict_segmentation()	6
2.1.2.7 remap_mask_labels()	7
2.1.2.8 resize_image()	7
2.1.2.9 resize_mask()	8
2.1.2.10 save_metrics_as_image()	8
2.1.2.11 save_summary_metrics_as_image()	9
2.1.2.12 train_naive_bayes_memory_efficient()	10
2.1.2.13 visualize_results()	11
Index	13

Chapter 1

File Index

1.1 File List

Here is a list of all documented files with brief descriptions:

NaiveBayesClassifier.py	
Optic disc and cup segmentation using Naive Bayes classifier	3

Chapter 2

File Documentation

2.1 NaiveBayesClassifier.py File Reference

Optic disc and cup segmentation using Naive Bayes classifier.

Functions

- [NaiveBayesClassifier.resize_image](#) (image, max_size=512)
Resize regular fundus image to reduce memory usage.
- [NaiveBayesClassifier.resize_mask](#) (mask, max_size=512)
Resize segmentation mask using nearest neighbor interpolation.
- [NaiveBayesClassifier.remap_mask_labels](#) (mask)
Remap mask pixel values to standardized label values.
- [NaiveBayesClassifier.extract_features_with_coords](#) (image, normalize_coords=True, use_both_colorspaces=True)
Extract color and spatial features from fundus image.
- [NaiveBayesClassifier.train_naive_bayes_memory_efficient](#) (fundus_files, mask_files, use_both_colorspaces=True, batch_size=5, max_size=512)
Train Naive Bayes classifier using batch processing and incremental learning.
- [NaiveBayesClassifier.predict_segmentation](#) (clf, scaler, test_image, use_both_colorspaces=True)
Predict segmentation mask for a test image.
- [NaiveBayesClassifier.calculate_dice_coefficient](#) (pred, gt, label)
Calculate Dice coefficient for a specific segmentation class.
- [NaiveBayesClassifier.calculate_iou](#) (pred, gt, label)
Calculate Intersection over Union (IoU) for a specific segmentation class.
- [NaiveBayesClassifier.evaluate_segmentation](#) (pred_mask, gt_mask)
Evaluate segmentation performance using multiple metrics.
- [NaiveBayesClassifier.save_metrics_as_image](#) (metrics_dict, image_name, save_path)
Save evaluation metrics as a formatted table image.
- [NaiveBayesClassifier.save_summary_metrics_as_image](#) (all_metrics, save_path)
Save overall summary metrics across all test images as a table image.
- [NaiveBayesClassifier.visualize_results](#) (original, predicted_mask, gt_mask=None, save_path=None)
Visualize segmentation results with side-by-side comparison.
- [NaiveBayesClassifier.main](#) ()
Memory-efficient main execution pipeline for optic disc and cup segmentation.

2.1.1 Detailed Description

Optic disc and cup segmentation using Naive Bayes classifier.

This module implements a memory-efficient pipeline for retinal image segmentation using a Gaussian Naive Bayes classifier with spatial and color features to segment optic disc, cup, and background regions from fundus images.

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Date

September 23, 2025

2.1.2 Function Documentation

2.1.2.1 `calculate_dice_coefficient()`

```
NaiveBayesClassifier.calculate_dice_coefficient (
    pred,
    gt,
    label)
```

Calculate Dice coefficient for a specific segmentation class.

Computes the Dice similarity coefficient (F1 score) between predicted and ground truth segmentation for a single class label. The Dice coefficient is calculated as: $2 * |X \cap Y| / (|X| + |Y|)$

Parameters

<i>pred</i>	Predicted segmentation mask.
<i>gt</i>	Ground truth segmentation mask.
<i>label</i>	Class label value to evaluate (0, 128, or 255).

Returns

Dice coefficient value in range [0, 1] where 1 indicates perfect overlap.

Calculate Dice coefficient for a specific class

2.1.2.2 `calculate_iou()`

```
NaiveBayesClassifier.calculate_iou (
    pred,
    gt,
    label)
```

Calculate Intersection over Union (IoU) for a specific segmentation class.

Computes the IoU (Jaccard index) between predicted and ground truth segmentation for a single class label. IoU is calculated as: $|X \cap Y| / |X \cup Y|$

Parameters

<i>pred</i>	Predicted segmentation mask.
<i>gt</i>	Ground truth segmentation mask.
<i>label</i>	Class label value to evaluate (0, 128, or 255).

Returns

IoU value in range [0, 1] where 1 indicates perfect overlap.

Calculate Intersection over Union (IoU) for a specific class

2.1.2.3 evaluate_segmentation()

```
NaiveBayesClassifier.evaluate_segmentation (
    pred_mask,
    gt_mask)
```

Evaluate segmentation performance using multiple metrics.

Computes comprehensive evaluation metrics including overall pixel accuracy, per-class Dice coefficients and IoU scores for all three classes (disc, cup, background), and their mean values.

Parameters

<i>pred_mask</i>	Predicted segmentation mask.
<i>gt_mask</i>	Ground truth segmentation mask.

Returns

Dictionary containing the following metrics:

- 'accuracy': Overall pixel accuracy
- 'dice_disc': Dice coefficient for disc class (0)
- 'dice_cup': Dice coefficient for cup class (128)
- 'dice_background': Dice coefficient for background class (255)
- 'dice_mean': Mean Dice across all classes
- 'iou_disc': IoU for disc class (0)
- 'iou_cup': IoU for cup class (128)
- 'iou_background': IoU for background class (255)
- 'iou_mean': Mean IoU across all classes

Evaluate segmentation performance

2.1.2.4 extract_features_with_coords()

```
NaiveBayesClassifier.extract_features_with_coords (
    image,
    normalize_coords = True,
    use_both_colorspaces = True)
```

Extract color and spatial features from fundus image.

Extracts pixel-wise features including RGB and optionally HSV color information along with normalized or absolute spatial coordinates for training or prediction.

Parameters

<i>image</i>	Input fundus image (BGR format).
<i>normalize_coords</i>	If True, normalize coordinates to [0,1] range. Default is True.
<i>use_both_colorspaces</i>	If True, extract both RGB and HSV features. Default is True.

Returns

Feature array of shape (height*width, n_features) where n_features is 8 (RGB + HSV + x + y) if use_both_colorspaces=True, or 5 (RGB + x + y) if use_both_colorspaces=False.

Extract color and spatial features from fundus image

2.1.2.5 main()

```
NaiveBayesClassifier.main ()
```

Memory-efficient main execution pipeline for optic disc and cup segmentation.

Executes the complete segmentation workflow including:

- Configuration of data directories and processing parameters
- Loading and preprocessing training data with memory-efficient batching
- Training Gaussian Naive Bayes classifier with incremental learning
- Testing on validation images with ground truth evaluation
- Computing comprehensive metrics (accuracy, Dice, IoU)
- Saving visualizations and metric tables for individual images
- Generating overall summary statistics across all test images

The pipeline uses memory optimization techniques including image resizing, batch processing, and explicit garbage collection to handle large datasets.

Returns

None. All outputs (predictions, metrics, visualizations) are saved to OUTPUT_DIR.

Memory-efficient main execution pipeline

2.1.2.6 predict_segmentation()

```
NaiveBayesClassifier.predict_segmentation (
    clf,
    scaler,
    test_image,
    use_both_colorspaces = True)
```

Predict segmentation mask for a test image.

Extracts features from the test image, standardizes them using the fitted scaler, and predicts pixel-wise class labels using the trained Naive Bayes classifier.

Parameters

<i>clf</i>	Trained GaussianNB classifier.
<i>scaler</i>	Fitted StandardScaler for feature normalization.
<i>test_image</i>	Input test fundus image (BGR format).
<i>use_both_colorspaces</i>	If True, use both RGB and HSV features. Default is True.

Returns

Predicted segmentation mask of shape (height, width) with values in {0, 128, 255}.

```
Predict segmentation mask for a test image
```

2.1.2.7 remap_mask_labels()

```
NaiveBayesClassifier.remap_mask_labels (
    mask)
```

Remap mask pixel values to standardized label values.

Converts grayscale mask values to exactly three discrete labels based on intensity thresholds. Handles cases where resizing creates intermediate values.

- Values 0-63 map to 0 (disc)
- Values 64-191 map to 128 (cup)
- Values 192-255 map to 255 (background)

Parameters

<i>mask</i>	Input grayscale mask with arbitrary intensity values.
-------------	---

Returns

Remapped mask with values in {0, 128, 255}.

```
Remap mask values to exactly 0, 128, or 255
Handles cases where resizing creates intermediate values
```

2.1.2.8 resize_image()

```
NaiveBayesClassifier.resize_image (
    image,
    max_size = 512)
```

Resize regular fundus image to reduce memory usage.

Proportionally resizes an image so that its largest dimension does not exceed the specified maximum size while maintaining aspect ratio using area interpolation.

Parameters

<i>image</i>	Input image array (BGR format).
<i>max_size</i>	Maximum dimension size in pixels. Default is 512.

Returns

Resized image or original image if already within size limit.

```
Resize regular image (fundus) to reduce memory usage
```

2.1.2.9 `resize_mask()`

```
NaiveBayesClassifier.resize_mask (
    mask,
    max_size = 512)
```

Resize segmentation mask using nearest neighbor interpolation.

Resizes a mask image while preserving discrete label values by using nearest neighbor interpolation to avoid creating intermediate values.

Parameters

<i>mask</i>	Input mask array with discrete labels.
<i>max_size</i>	Maximum dimension size in pixels. Default is 512.

Returns

Resized mask or original mask if already within size limit.

```
Resize mask using nearest neighbor to preserve discrete labels
```

2.1.2.10 `save_metrics_as_image()`

```
NaiveBayesClassifier.save_metrics_as_image (
    metrics_dict,
    image_name,
    save_path)
```

Save evaluation metrics as a formatted table image.

Creates and saves a styled table displaying all evaluation metrics including pixel accuracy, Dice coefficients, and IoU scores for a single test image. The table includes color-coded rows and highlighted mean scores.

Parameters

<i>metrics_dict</i>	Dictionary containing evaluation metrics from <code>evaluate_segmentation()</code> .
<i>image_name</i>	Name of the image being evaluated (used in title).
<i>save_path</i>	Full file path where the metrics table image will be saved.

Returns

None. Saves the metrics table as a PNG image to the specified path.

Save evaluation metrics as a table image

```
Parameters:
-----
metrics_dict : dict
    Dictionary containing evaluation metrics
image_name : str
    Name of the image being evaluated
save_path : str
    Full path where to save the metrics image
```

2.1.2.11 save_summary_metrics_as_image()

```
NaiveBayesClassifier.save_summary_metrics_as_image (
    all_metrics,
    save_path)
```

Save overall summary metrics across all test images as a table image.

Computes mean and standard deviation for all metrics across multiple test images and saves a formatted table displaying the aggregated statistics.

Parameters

<i>all_metrics</i>	List of metrics dictionaries from all test images. Each dictionary should have the same keys as returned by <code>evaluate_segmentation()</code> .
<i>save_path</i>	Full file path where the summary metrics table image will be saved.

Returns

None. Saves the summary metrics table as a PNG image. Returns early if `all_metrics` is empty.

Save overall summary metrics as a table image

```
Parameters:
-----
all_metrics : list of dict
    List of metrics dictionaries from all test images
save_path : str
    Path to save the summary image
```

2.1.2.12 `train_naive_bayes_memory_efficient()`

```
NaiveBayesClassifier.train_naive_bayes_memory_efficient (
    fundus_files,
    mask_files,
    use_both_colorspaces = True,
    batch_size = 5,
    max_size = 512)
```

Train Naive Bayes classifier using batch processing and incremental learning.

Trains a Gaussian Naive Bayes classifier on fundus images and their segmentation masks using incremental learning with batch processing to reduce memory usage. Uses `partial_fit` for online learning across batches.

Parameters

<i>fundus_files</i>	List of file paths to training fundus images.
<i>mask_files</i>	List of file paths to corresponding segmentation masks.
<i>use_both_colorspaces</i>	If True, use both RGB and HSV color spaces. Default is True.
<i>batch_size</i>	Number of images to process per batch. Default is 5.
<i>max_size</i>	Maximum image dimension for resizing. Default is 512.

Returns

Tuple of (trained_classifier, fitted_scaler) where:

- `trained_classifier` is a `GaussianNB` object trained on all batches
- `fitted_scaler` is a `StandardScaler` for feature normalization

Memory-efficient training with batching and incremental learning

Parameters:

```
fundus_files : list
    List of paths to fundus images
mask_files : list
    List of paths to mask images
use_both_colorspaces : bool
    Whether to use both RGB and HSV features
batch_size : int
    Number of images to process at once
max_size : int
    Maximum image dimension for resizing
```

Returns:

```
clf : GaussianNB
    Trained classifier
scaler : StandardScaler
    Fitted feature scaler
```

2.1.2.13 visualize_results()

```
NaiveBayesClassifier.visualize_results (
    original,
    predicted_mask,
    gt_mask = None,
    save_path = None)
```

Visualize segmentation results with side-by-side comparison.

Creates a visualization comparing the original image, predicted segmentation mask, and optionally the ground truth mask. Saves the visualization as an image file.

Parameters

<i>original</i>	Original fundus image (BGR format).
<i>predicted_mask</i>	Predicted segmentation mask with values {0, 128, 255}.
<i>gt_mask</i>	Ground truth mask (optional). If provided, creates 3-panel visualization. If None, creates 2-panel visualization. Default is None.
<i>save_path</i>	File path to save the visualization image. If None, the figure is only closed without saving. Default is None.

Returns

None. Either saves the visualization to disk or closes the figure.

Visualize segmentation results

Index

- calculate_dice_coefficient
 - NaiveBayesClassifier.py, [4](#)
- calculate_iou
 - NaiveBayesClassifier.py, [4](#)
- evaluate_segmentation
 - NaiveBayesClassifier.py, [5](#)
- extract_features_with_coords
 - NaiveBayesClassifier.py, [5](#)
- main
 - NaiveBayesClassifier.py, [6](#)
- NaiveBayesClassifier.py, [3](#)
 - calculate_dice_coefficient, [4](#)
 - calculate_iou, [4](#)
 - evaluate_segmentation, [5](#)
 - extract_features_with_coords, [5](#)
 - main, [6](#)
 - predict_segmentation, [6](#)
 - remap_mask_labels, [7](#)
 - resize_image, [7](#)
 - resize_mask, [8](#)
 - save_metrics_as_image, [8](#)
 - save_summary_metrics_as_image, [9](#)
 - train_naive_bayes_memory_efficient, [9](#)
 - visualize_results, [10](#)
- predict_segmentation
 - NaiveBayesClassifier.py, [6](#)
- remap_mask_labels
 - NaiveBayesClassifier.py, [7](#)
- resize_image
 - NaiveBayesClassifier.py, [7](#)
- resize_mask
 - NaiveBayesClassifier.py, [8](#)
- save_metrics_as_image
 - NaiveBayesClassifier.py, [8](#)
- save_summary_metrics_as_image
 - NaiveBayesClassifier.py, [9](#)
- train_naive_bayes_memory_efficient
 - NaiveBayesClassifier.py, [9](#)
- visualize_results
 - NaiveBayesClassifier.py, [10](#)